EMPIRICALLY DERIVED CRITERION WEIGHTS FOR WORKER RANKING IN AN INFORMAL HOUSEHOLD SERVICES MARKETPLACE

Individual component: Intelligent Worker Recommendation Engine

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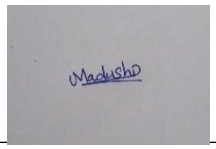
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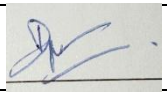
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ABSTRACT

Online marketplaces for informal household maintenance services must rank available workers in response to each customer request. The ranking depends on several competing criteria - formal qualification, customer reputation, accumulated experience, and geographic proximity - and no single criterion identifies the most suitable worker. Existing platforms aggregate these criteria using weights assigned by system designers, and no empirically derived weight set has been published for this sector in Sri Lanka. This thesis addresses that gap.

A profile-based survey instrument was constructed containing twelve worker profiles in which the four criteria were varied orthogonally, with pairwise correlations not exceeding 0.208. The instrument was administered to forty-one hiring professionals including contractors, site supervisors, service company owners, and human resource managers. After removing two non-discriminating responses, thirty-nine valid respondents produced 468 suitability evaluations. Ordinary least squares regression on standardised variables produced criterion weights of 0.396 for proximity, 0.305 for customer rating, 0.215 for experience, and 0.084 for National Vocational Qualification level. Three of the four coefficients reached significance at $p < 0.001$; the qualification coefficient did not ($p = 0.092$). The model explains 20.0 per cent of variance.

The weights were embedded in a four-stage pipeline comprising eligibility filtering, min-max normalisation, weighted aggregation, and rank selection, deployed as a REST service. The system ranks ten thousand workers in under five milliseconds, produces scores bounded within zero to one hundred, behaves monotonically in all four criteria, returns identical orderings across repeated executions, and decomposes every score into its exact criterion contributions.

A secondary instrument collecting direct importance rankings from the same respondents inverted the regression ordering, placing proximity last where behavioural evaluation placed it first. The divergence, at a Spearman correlation of -0.400, remained stable across four independent rank-to-weight conversion procedures. This evidences a stated-versus-revealed preference gap in worker selection and demonstrates the risk of parameterising recommendation systems from self-reported importance.

Keywords: multi-criteria decision making, recommender systems, weighted sum model, revealed preference, service marketplaces

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I thank the forty-one industry professionals - contractors, site supervisors, service company owners, and human resource managers - who gave their time to complete the survey instrument. Their professional judgement forms the empirical foundation of this thesis, and the work would not have been possible without their participation.

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
AHP	Analytic Hierarchy Process
API	Application Programming Interface
GPS	Global Positioning System
MCDM	Multi-Criteria Decision Making
NVQ	National Vocational Qualification
OLS	Ordinary Least Squares
REST	Representational State Transfer
ROC	Rank Order Centroid
RPL	Recognition of Prior Learning
SEM	Structural Equation Modelling
TOPSIS	Technique for Order Preference by Similarity to Ideal Solution
TVEC	Tertiary and Vocational Education Commission
WSM	Weighted Sum Model

1 INTRODUCTION

1.1 Overall project context

The group project develops an artificial intelligence enabled services marketplace for household maintenance in Sri Lanka. The platform connects homeowners requiring plumbing, electrical, carpentry, masonry, and painting services with skilled workers available in their area. The complete system comprises four interacting components. Component 01 interprets a customer's free-text service request and resolves it to a structured job category. Component 02, the subject of this thesis, ranks available workers against that request and returns an ordered shortlist. Component 03 processes customer reviews and produces a trust rating for each worker. Component 04 predicts the expected price of the requested job.

The sector this platform addresses remains largely unmediated in Sri Lanka. Household maintenance work is typically arranged through personal referral, with customers contacting several workers in sequence before securing one who is qualified, available, and willing to travel. The absence of a structured intermediary produces inefficiency on both sides of the market: customers expend effort searching, and competent workers without established referral networks remain underutilised.

1.2 Individual component context

Component 02 occupies the decision layer of the platform. It receives a structured job category from Component 01 together with the customer's current location, and it draws worker attributes from the platform database and trust ratings from Component 03. Its output is an ordered list of the twenty most suitable workers for that request.

The component is the point at which the platform's commitments to fairness and transparency are operationalised. The ordering it produces determines which workers become visible to a customer and therefore which workers receive work. A ranking function that systematically disadvantages a class of workers - those newly registered, those located outside the immediate area, or those holding qualifications the system does not recognize - produces a material economic effect on those individuals. The component must therefore be not only accurate but auditable.

1.3 Problem statement

When a customer requests a service, the platform must determine which of the available workers is most suitable. Suitability is not directly observable. It is inferred from measurable worker attributes, and those attributes frequently conflict.

Consider six plumbers, all available at the moment of request. The first holds the highest National Vocational Qualification level but carries the weakest customer rating. The second is the closest to the customer but holds the lowest qualification. The third has a perfect rating but is the most distant. No worker dominates the others on every criterion, and consequently no single-criterion sort identifies the correct answer. The problem is one of multi-criteria aggregation: the criteria must be combined into a single comparable score, and that combination requires weights.

The central difficulty is that the weights are unknown. Assigning them by designer judgement produces a system whose behaviour reflects the designer's assumptions rather than the domain's reality, and such a system cannot be defended when challenged.

1.4 Research gap

Three gaps motivate this work.

First, no empirically derived criterion weight set has been published for worker selection in Sri Lanka's informal household services sector. Practitioners in this sector make selection decisions routinely, but the relative importance they assign to qualification, reputation, experience, and proximity has not been measured or documented.

Second, existing marketplace platforms aggregate criteria using weights that are either undisclosed or assigned heuristically. Where weighting methodology is reported in the recommender systems literature, it typically relies on the Analytic Hierarchy Process with pairwise comparisons supplied by the researchers, which relocates rather than removes the subjectivity.

Third, the relationship between what practitioners state is important and what their actual evaluations reveal has not been examined in this domain. If the two diverge,

systems parameterised from self-reported importance will be systematically misweighted.

1.5 Aim and objectives

The aim of this research is to design, derive the parameters for, implement, and validate a worker ranking component whose criterion weights are established empirically from the judgement of domain practitioners rather than assigned by the researcher.

The supporting objectives are as follows.

Objective 1. To identify the criteria that determine worker suitability in the household maintenance domain and establish which of these are reliably available to the platform at the time of a request.

Objective 2. To design a survey instrument capable of eliciting suitability judgements from which criterion weights can be recovered statistically, with the criteria varied orthogonally so that individual effects are separable.

Objective 3. To administer the instrument to practising hiring professionals and derive criterion weights by regression on the resulting evaluations.

Objective 4. To design and implement a ranking pipeline that applies the derived weights and produces scores that are bounded, monotonic, deterministic, and decomposable.

Objective 5. To validate the component through functional testing, performance measurement, and cross-comparison of the derived weights against an independent elicitation method.

Objective 6. To deploy the component as a network service that the remaining components and the mobile application can consume.

1.6 Research questions or hypotheses

RQ1. What relative weights do practising hiring professionals assign to National Vocational Qualification level, customer rating, years of experience, and geographic proximity when judging worker suitability?

RQ2. Can these weights be recovered reliably by regression on profile-based suitability evaluations, and do the resulting coefficients reach statistical significance?

RQ3. Does the weighting recovered from behavioural evaluation agree with the weighting obtained when the same professionals are asked to rank the criteria by importance directly?

H1. All four criteria contribute significantly to judged worker suitability.

Hypothesis H1 was only partially supported. Three criteria reached significance; National Vocational Qualification level did not. The consequence for the design is examined in Section 7.1.1.

H2. The criterion ordering recovered by regression agrees with the ordering obtained by direct importance ranking.

Hypothesis H2 was not supported, and the nature of the disagreement forms a substantive finding of this work, reported in Section 6.1.3 and interpreted in Section 7.1.2.

1.7 Scope limitations and assumptions

Scope. The component addresses ranking only. It does not perform category resolution, which belongs to Component 01, nor sentiment analysis of reviews, which belongs to Component 03, nor price estimation, which belongs to Component 04. Ranking operates within a single job category; cross-category comparison is explicitly excluded, since a worker's competence as a plumber carries no information about their competence as an electrician.

Limitations. The survey sample was drawn from Sri Lanka, predominantly the Western Province, and the derived weights should not be assumed to transfer to other regions or markets. Respondent identity could not be independently verified because the instrument was administered anonymously. The regression explains 20.0 per cent of variance in suitability ratings; the remainder reflects heterogeneity between individual professionals that a single weight set cannot capture.

Assumptions. It is assumed that the twelve profiles presented in the survey adequately span the attribute space encountered in practice, that respondents answered according

to their professional judgement rather than strategically, and that the four criteria available to the platform are sufficient to produce a useful ranking. It is further assumed that National Vocational Qualification level is a meaningful competence signal in this domain, which is supported by the Tertiary and Vocational Education Commission's Recognition of Prior Learning pathway described in Section 2.2.2.

1.8 Individual contribution

My individual contribution comprises the following.

The design of the profile-based survey instrument, including the construction of twelve orthogonally varied worker profiles and the verification that criterion correlations across the profile set remain below 0.21.

The administration of the instrument to forty-one industry professionals, the cleaning of the resulting dataset, and the reshaping of responses into the long-format structure required for regression.

The derivation of criterion weights by ordinary least squares regression on 468 standardised observations, and the independent derivation of weights from the direct-ranking instrument by four established rank-to-weight conversion procedures.

The design and implementation of the four-stage ranking pipeline, including the decision to treat distance as an inverted cost criterion rather than a subtractive penalty, and the cold-start mechanism that keeps newly registered workers discoverable.

The functional validation of the implementation against boundedness, monotonicity, determinism, and robustness criteria, and the performance characterisation across candidate pool sizes from one hundred to fifty thousand.

The deployment of the component as a REST service with an explainability endpoint that decomposes any score into its criterion contributions.

1.9 Thesis structure

Chapter 2 reviews multi-criteria decision making methods, weight elicitation techniques, and the stated-versus-revealed preference literature, and synthesises the gap this work addresses. Chapter 3 presents the research methodology, covering the

survey design, participants, analytical procedure, and ethical considerations. Chapter 4 presents the component design, including requirements, architecture, design alternatives considered, and the detailed scoring model. Chapter 5 describes the implementation, integration with the remaining components, verification testing, and deployment. Chapter 6 reports the research and technical results. Chapter 7 interprets the findings, states the contributions, and examines threats to validity. Chapter 8 addresses commercialisation and sustainability. Chapter 9 concludes and identifies future directions.

2 LITERATURE REVIEW AND RELATED WORK

2.1 Review method and scope

The review was conducted across IEEE Xplore, ScienceDirect, SpringerLink, and the ACM Digital Library, supplemented by searches of the Google Scholar index for grey literature and regional publications not indexed in the primary databases. Search terms combined the domain concepts “worker recommendation”, “service marketplace”, “gig platform”, and “labour matching” with the method concepts “multi-criteria decision making”, “weighted sum”, “criterion weight”, “weight elicitation”, and “revealed preference”.

Sources were included where they addressed the aggregation of multiple criteria into a ranking, the elicitation of criterion weights, or the measurement of preference in a hiring or selection context. Sources were excluded where the ranking problem was single-criterion, where the method required training data unavailable in a pre-launch platform, or where the application domain was sufficiently distant that the criteria did not transfer.

Three bodies of literature proved relevant and are reviewed in turn: multi-criteria decision making methods, weight elicitation procedures, and the economic literature on stated versus revealed preference.

2.2 Domain and theoretical foundations

2.2.1 Multi-criteria decision making

Multi-criteria decision making addresses the selection or ranking of alternatives evaluated against several criteria that are typically incommensurable and frequently in conflict. The field distinguishes between benefit criteria, for which higher values are preferable, and cost criteria, for which lower values are preferable. Because criteria are measured in different units, aggregation requires normalisation to a common scale before any combination is performed.

The Weighted Sum Model is the oldest and most widely applied aggregation method. It computes a composite score as the weighted linear combination of normalised

criterion values. Its principal virtue is transparency: the contribution of each criterion to the final score is explicit and separable. Its principal limitation is compensability, meaning that strong performance on one criterion can fully offset weak performance on another.

The Technique for Order Preference by Similarity to Ideal Solution, introduced by Hwang and Yoon, ranks alternatives by their relative closeness to a hypothetical ideal solution and distance from a hypothetical anti-ideal solution. It handles trade-offs differently from the weighted sum, penalising alternatives that are extreme on any single criterion. It does not, however, produce weights; it consumes them, and therefore requires a separate elicitation procedure.

2.2.2 The Sri Lankan National Vocational Qualification framework

The National Vocational Qualifications Framework of Sri Lanka, administered by the Tertiary and Vocational Education Commission, defines a seven-level structure in which qualifications are assigned according to the complexity of the processes a holder performs, the learning demand of those processes, and the responsibility the holder carries. Levels one and two represent pre-vocational stages; level three is the minimum qualification associated with independent trade work, and levels five and above correspond to technician, supervisory, and professional roles.

Of particular relevance to this component is the Recognition of Prior Learning pathway. The framework permits an experienced worker with no formal training to obtain certification by assessment of demonstrated competence, through practical demonstration, interview, and verification of work history. A worker with approximately eighteen months of verifiable industry experience may be certified at level two or three by this route. The qualification therefore certifies competence rather than classroom attendance, which is the property that makes it suitable as a ranking criterion: a skilled but formally untrained worker is not excluded from the framework but is routed through a different assessment pathway.

2.3 Related research

Research on recommendation in labour platforms divides broadly into matching approaches and ranking approaches. Matching approaches treat the problem as one of correspondence between a job specification and a worker profile, typically using text similarity or embedding distance over skill descriptions. These approaches are effective where job requirements are expressed in rich natural language, as in freelance software platforms, but transfer poorly to household maintenance, where the job category is coarse and the discriminating factors are attributes rather than skills.

Ranking approaches treat the problem as one of ordering a candidate set by composite suitability. Work in this tradition has applied collaborative filtering where interaction history is available, and multi-criteria aggregation where it is not. The cold-start condition - a new platform with no interaction history, or a new worker with no completed jobs - is widely identified as the principal obstacle to collaborative approaches and is the condition under which this platform launches.

A smaller body of work addresses competency-based evaluation, in which worker capability is assessed against a structured framework rather than inferred from outcomes. This literature supports the inclusion of a formal qualification measure as a ranking criterion, though it does not establish how heavily such a measure should be weighted relative to reputation or convenience factors.

2.4 Existing systems products and techniques

Commercial service marketplaces operating in and near the Sri Lankan market expose ranking behaviour that can be inferred from their interfaces. Ride-hailing and delivery platforms rank predominantly by proximity, with acceptance rate and rating serving as secondary filters. General task marketplaces rank by a combination of star rating, completed job count, and responsiveness. In neither category is formal vocational qualification used as a ranking input, and in neither category is the weighting disclosed.

Three limitations are common to these systems. The weighting is opaque, so a worker cannot determine why they ranked where they did. Newly registered workers are

structurally disadvantaged, since the criteria that drive ranking are accumulated through work that ranking determines access to. And formal competence signals are absent, which in a market where a national qualification framework exists represents an unused source of information.

2.5 Gap synthesis and conceptual framework

The literature establishes that multi-criteria aggregation is the appropriate method class for this problem, that the weighted sum offers transparency at the cost of compensability, and that weights must be obtained from some external procedure. It does not supply weights for this domain, and the available elicitation procedures either depend on researcher judgement or reduce the problem to an ordinal ranking.

This work addresses the gap by eliciting weights from the behavioural evaluations of domain practitioners. The conceptual framework treats suitability as a latent construct that cannot be measured directly but which determines observable evaluation. Four measurable attributes act as causes of suitability, and professionals' suitability judgements serve as the observed indicator from which the causal weights are recovered. This formulation follows the structural logic of structural equation modelling, though the recovery procedure used here is ordinary least squares regression on a single observed indicator rather than a full latent variable model, which the sample size does not support.

3 RESEARCH METHODOLOGY

3.1 Research design

The study follows a quantitative, cross-sectional design in which criterion weights are recovered by regression from profile-based evaluations. The design is observational rather than experimental in the sense that no treatment is administered, but it borrows from conjoint analysis the practice of systematically varying attributes across a profile set so that individual attribute effects can be isolated statistically.

Two elicitation instruments were administered to the same respondents. The primary instrument is behavioural: respondents evaluated concrete worker profiles and their judgements were decomposed into implicit criterion weights. The secondary instrument is attitudinal: respondents ranked the criteria by importance directly. The two were compared to assess convergent validity, and their divergence became a finding in its own right.

3.2 Data participants or study materials

3.2.1 Sampling frame and recruitment

The target population comprised individuals in Sri Lanka whose professional role involves hiring, assigning, or supervising household maintenance workers. Purposive sampling was used. The survey link was not published publicly; it was distributed directly to identified organisations and individuals by electronic mail and messaging application, accompanied by an authorisation letter from the research supervisor on institutional letterhead. A distribution log recording recipient organisation, contact, date, and channel was maintained throughout.

Forty-one responses were received. Two were removed because the respondent assigned an identical rating to all twelve profiles, indicating no discrimination between profiles and therefore no usable information for regression. Thirty-nine valid responses remained.

3.2.2 Respondent characteristics

Table 3.1 reports the professional role distribution of valid respondents.

Table 3.1: Respondent professional role distribution

Professional role	n
Service company owner or manager	11
Contractor or site supervisor	8
Human resource manager or executive	6
Other trade-related roles	14
Total	39

Table 3.2 reports hiring experience. One respondent did not supply a classifiable response to this item.

Table 3.2: Respondent hiring experience distribution

Years involved in hiring or assigning workers	n
Less than 2 years	23
2 to 5 years	8
6 to 10 years	4
More than 10 years	3
Total classified	38

3.2.3 Instrument design

The instrument comprised three sections. Section A collected professional role, hiring experience, and trade categories, serving as an eligibility screen. Section B presented twelve worker profiles for suitability evaluation. Section C collected a direct importance ranking of the four criteria.

Each Section B profile specified a National Vocational Qualification level, a customer rating out of five, years of experience, and distance in kilometres. Respondents rated suitability on a ten-point scale anchored at one, indicating not suitable at all, and ten, indicating extremely suitable. A ten-point scale was selected over a five-point scale for two reasons: with twelve profiles and only five response levels, many profiles would receive identical ratings and the regression would lose discriminating variance;

and because the profiles themselves display a five-point customer rating, a five-point suitability scale risked confusion between the stimulus and the response.

Table 3.3 presents the profile design matrix.

Table 3.3: Worker profile design matrix

Profile	NVQ level	Customer rating	Experience (years)	Distance (km)
1	7	4.8	12	2.0
2	3	4.9	2	1.5
3	5	3.2	8	3.0
4	6	4.5	5	18.0
5	4	4.0	15	6.0
6	7	3.0	3	12.0
7	3	3.5	10	20.0
8	6	5.0	1	4.0
9	5	4.7	14	22.0
10	4	3.8	6	1.0
11	7	4.2	9	8.0
12	3	4.4	4	15.0

National Vocational Qualification levels were restricted to the range three to seven. Levels one and two were excluded on the grounds that they represent pre-vocational stages and workers holding them would not undertake independent trade work through the platform.

The profiles were constructed so that the four criteria vary independently of one another. Profile 2 pairs a low qualification with an excellent rating; Profile 6 pairs the highest qualification with a poor rating; Profile 8 pairs a strong qualification and perfect rating with minimal experience. This construction forces respondents to make genuine trade-offs rather than responding to a single dominant attribute, and it is the property that allows regression to attribute variance to individual criteria rather than to their correlation.

Table 3.4 reports the resulting correlation structure.

Table 3.4: Criterion orthogonality across the profile set

	NVQ	Rating	Experience	Distance
NVQ	1.000	0.008	0.012	−0.103
Rating	0.008	1.000	−0.067	−0.077
Experience	0.012	−0.067	1.000	0.208
Distance	−0.103	−0.077	0.208	1.000

The largest absolute off-diagonal correlation is 0.208, between experience and distance. This is well below thresholds at which multicollinearity distorts coefficient estimation, and the design is therefore adequate for the recovery of independent criterion effects.

Section C presented the four criteria in a forced-choice ranking grid with four ordinal positions, configured so that each position could be assigned only once. This constraint was enforced by the survey platform, eliminating the possibility of tied or incomplete rankings.

3.3 Experimental or study procedure

Responses were collected through an online form over the data collection period. Each respondent completed Section A, then evaluated the twelve Section B profiles in fixed order, then completed the Section C ranking. All items were marked as required, since a missing profile evaluation removes an entire observation from the regression rather than a single data point.

Collected responses were exported and screened. Responses exhibiting zero standard deviation across the twelve profile ratings were removed as non-discriminating. The remaining responses were reshaped from wide format, in which each respondent occupies one row with twelve rating columns, to long format, in which each respondent-profile pair occupies one row carrying the four criterion values and the assigned suitability rating. Thirty-nine respondents evaluating twelve profiles produced 468 observations.

3.4 Measures and evaluation metrics

Research measures. The dependent measure is the suitability rating assigned by a respondent to a profile, on a scale of one to ten. The independent measures are the four criterion values defining each profile. Model quality is assessed by the coefficient of determination, the overall F statistic, and the per-coefficient t statistic and associated p value.

Agreement measure. Agreement between the regression-derived ordering and the direct-ranking ordering is quantified by Spearman's rank correlation coefficient.

Technical measures. The implementation is evaluated against boundedness, meaning that scores remain within the interval zero to one hundred for all admissible and inadmissible inputs; monotonicity, meaning that increasing a benefit criterion does not decrease the score and increasing a cost criterion does not increase it; determinism, meaning that repeated execution on identical input produces identical ordering; and latency, measured as mean and ninety-fifth percentile wall-clock time across candidate pools of varying size.

3.5 Data analysis

Weights were derived by ordinary least squares regression. All variables, dependent and independent, were standardised to zero mean and unit variance before fitting. Standardisation is necessary because the four criteria are measured in different units and across different ranges; without it, the magnitude of a raw coefficient reflects the unit of measurement as much as the importance of the criterion, and coefficients cannot be compared.

The regression model is

$$z(S) = \beta_1 \cdot z(N) + \beta_2 \cdot z(R) + \beta_3 \cdot z(E) + \beta_4 \cdot z(D) + \varepsilon$$

where $z(\cdot)$ denotes standardisation, S is the suitability rating, and N, R, E, and D are National Vocational Qualification level, customer rating, experience, and distance respectively.

Criterion weights were computed from the standardised coefficients as

$$w_i = |\beta_i| \div \Sigma_{j=1}^{4} |\beta_j|$$

The absolute value is taken because the distance coefficient is negative by construction - greater distance reduces suitability - while the weight expresses the magnitude of the criterion's influence. The direction is preserved separately in the scoring function, where distance is treated as a cost criterion. Normalisation by the sum ensures the weights total unity.

Section C rankings were converted to weights by four established procedures so that the comparison with the regression result would not depend on any single conversion choice. The mean-rank inverse method weights each criterion by the reciprocal of its mean assigned rank, normalised. The Rank Sum method assigns weight proportional to $n - r_i + 1$ where r_i is the criterion's position and n the number of criteria. The Rank Reciprocal method assigns weight proportional to $1 \div r_i$. The Rank Order Centroid method assigns

$$w_i = (1 \div n) \cdot \Sigma_{j=ri}^{n} (1 \div j)$$

All analysis was conducted in Python using the pandas, statsmodels, and scipy libraries.

3.6 Ethics privacy security and responsible AI

Informed participation. The instrument opened with a statement of the research purpose, the estimated completion time, the voluntary nature of participation, and the intended use of the data. The supervisor's authorisation letter was supplied with every distribution.

Anonymity. No personally identifying information was collected. This was a deliberate design choice rather than an omission, made for two reasons: it honours the anonymisation commitment in the supervisor's letter, and it reduces social desirability bias, which is a material concern when professionals are asked to disclose how they weigh criteria such as convenience against competence.

The consequence is that individual respondent identity cannot be independently verified. Verification therefore operated at the distribution stage rather than the

collection stage. The survey link was not published publicly, it was sent to identified recipients, and a distribution log was maintained. Eligibility was further confirmed by the Section A screening items. This limitation is stated explicitly in Section 7.5.

Data handling. Responses were exported to a single working file, stored locally, and used solely for the analysis reported here. No response data was transmitted to third-party services.

Responsible system design. The component determines which workers become visible to customers and therefore influences access to income. Three responses to this follow. The scoring function is linear and its contributions are separable, so any ranking decision can be decomposed and explained to the affected worker. The cold-start mechanism described in Section 4.4.4 prevents newly registered workers from being permanently excluded by the absence of review history. And the dominant weight on proximity, which structurally disadvantages workers in peripheral areas, is reported openly rather than concealed, and is identified in Section 9.4 as requiring mitigation.

3.7 Reproducibility and research artefacts

The following artefacts support reproduction of the results reported in this thesis.

The cleaned long-format dataset of 468 observations, carrying respondent identifier, the four criterion values, and the assigned suitability rating.

The regression script, which reproduces the coefficients, standard errors, t statistics, p values, model fit statistics, and derived weights reported in Section 6.1.

The derived weight bundle, serialised to disk, containing the four weights, criterion directions, normalisation bounds, and the model metadata including sample size and fit statistics.

The ranking engine implementation and its test suite, covering boundedness, monotonicity, determinism, robustness, and performance.

The deployed service, which exposes the weights and metadata through a public endpoint so that the parameters in use can be verified against those reported here.

4 PRODUCT AND SOLUTION DESIGN

4.1 Requirements

4.1.1 Functional requirements

FR1. The component shall accept a job category, an optional worker type constraint, and a set of candidate worker records.

FR2. The component shall exclude from consideration any worker whose category does not match the request, whose availability status is not online, or whose type does not match an applied type constraint.

FR3. The component shall compute a suitability score for each eligible worker from National Vocational Qualification level, customer rating, years of experience, and distance from the customer.

FR4. The component shall return the eligible workers in descending order of score, truncated to a configurable limit defaulting to twenty.

FR5. The component shall, on request, return the per-criterion decomposition of each score.

FR6. The component shall apply a neutral rating to workers whose accumulated review count falls below a defined threshold.

4.1.2 Non-functional requirements

NFR1 Performance. A ranked list shall be returned within two seconds for the expected candidate pool size.

NFR2 Correctness. Scores shall lie within the closed interval zero to one hundred for all inputs, including inputs outside the defined criterion ranges.

NFR3 Monotonicity. Increasing a benefit criterion shall not decrease the score; increasing a cost criterion shall not increase it.

NFR4 Determinism. Repeated execution on identical input shall produce identical ordering.

NFR5 Explainability. Every score shall be decomposable into the contribution of each criterion.

NFR6 Maintainability. Criterion weights shall be externally configurable without modification of the scoring logic.

NFR7 Interoperability. The component shall be consumable over HTTP using JSON request and response payloads.

4.2 Component architecture within the complete solution

The component sits downstream of category resolution. Component 01 supplies the resolved job category. The platform database supplies worker records including qualification, experience, availability, and location. Component 03 supplies the trust rating for each worker. Component 02 consumes these inputs, produces the ranked shortlist, and passes it to the application layer for display. Price estimation for the requested job is produced independently by Component 04 and presented alongside the shortlist; the two components do not exchange data directly.

A deliberate architectural decision is that worker records arrive in the request rather than being retrieved by the component itself. The component holds no database connection and no knowledge of the persistence layer. This decoupling allows the data layer to be developed, changed, or replaced without modification to the ranking logic, and it allows the ranking logic to be tested in isolation with synthetic worker sets.

The internal structure is a four-stage pipeline. Stage one applies hard eligibility filters. Stage two normalises each criterion to a common scale. Stage three computes the weighted aggregate. Stage four sorts and truncates.

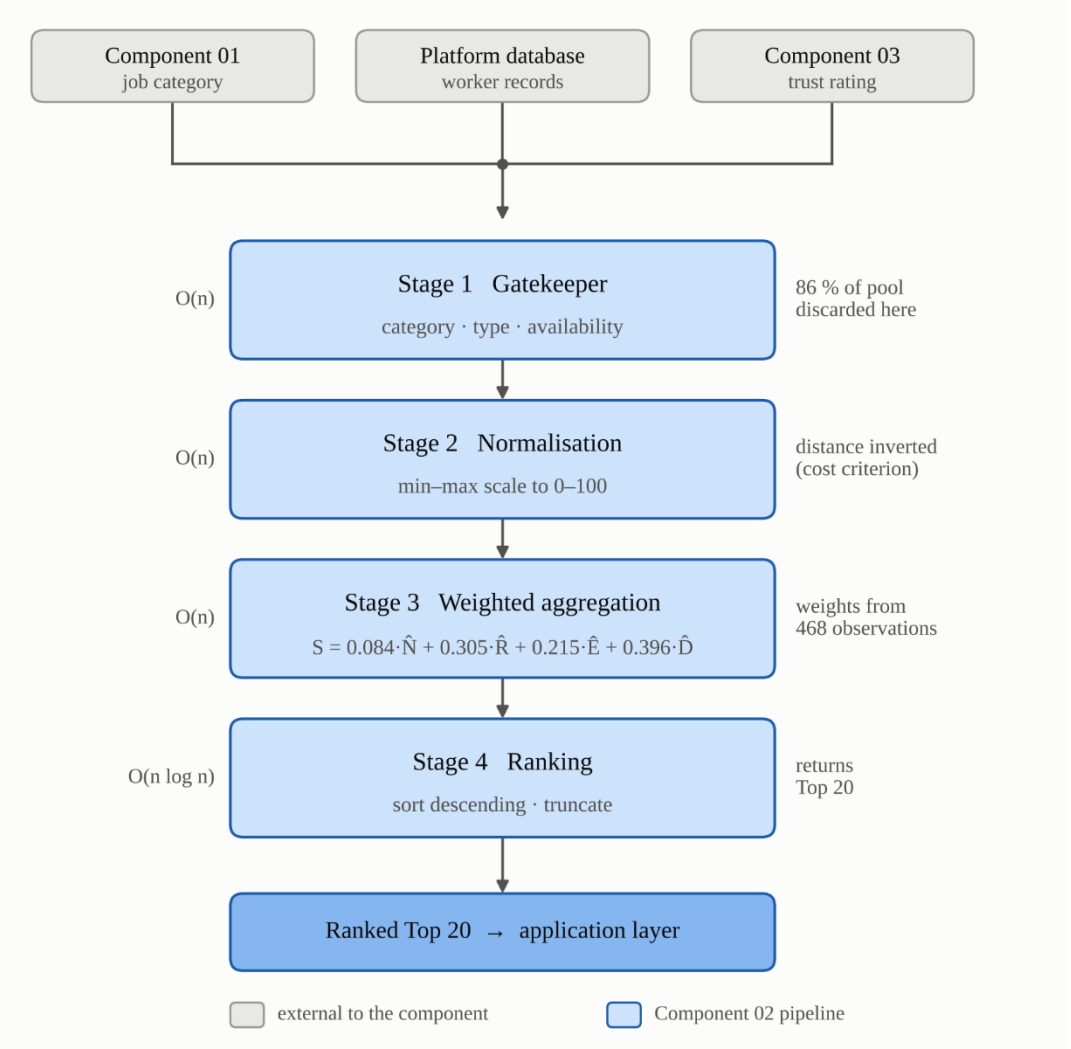


Figure 4.1: System flow of the four stage

4.3 Design alternatives and decisions

4.3.1 Choice of aggregation method

Three candidate approaches were considered.

A supervised learning model trained on historical booking outcomes would learn the ranking function from data. This was rejected because the platform has not launched and no outcome history exists, and because the resulting model would not be able to explain an individual ranking decision.

The Technique for Order Preference by Similarity to Ideal Solution would rank alternatives by relative closeness to an ideal solution. This was considered seriously and remains a reasonable alternative. It was not selected because it still requires externally supplied weights and therefore does not remove the central problem, and

because its scores are less directly interpretable than those of a weighted sum: a relative closeness coefficient does not decompose into criterion contributions in the way a linear score does.

The Weighted Sum Model was selected for its transparency. Its principal theoretical weakness, compensability, is acceptable in this application: a worker who is excellent on three criteria and weak on one should indeed rank above a worker who is mediocre on all four, and the hard filters in stage one already exclude workers who fail an absolute requirement.

4.3.2 Choice of weight elicitation method

An earlier iteration of this work derived weights by the Analytic Hierarchy Process, constructing a pairwise comparison matrix from positions reported in the literature, normalising it, and extracting weights as row means. That approach produced a consistency ratio of 0.035, within the accepted threshold of 0.10, and was internally valid.

It was abandoned on the grounds that the pairwise judgements, although informed by literature, were ultimately supplied by the researcher. A weight derived from a judgement the researcher made is not an empirical weight, however rigorous the arithmetic applied to it afterwards. The survey-based approach was adopted in its place because it sources the judgements from the practitioners who make these decisions professionally.

4.3.3 Treatment of the distance criterion

Distance is a cost criterion: lower values are preferable. Two treatments are available.

The first subtracts a weighted distance penalty from the sum of the benefit terms. This formulation is intuitive but admits negative scores when the distance term exceeds the sum of the others, and the resulting score has no fixed upper or lower bound.

The second inverts the criterion during normalisation, so that zero distance maps to one hundred and maximum distance maps to zero. The quantity entering the equation is then proximity rather than distance, all four terms are additive, and the score is bounded in the interval zero to one hundred by construction.

The second treatment was selected. It satisfies NFR2 without requiring a clamping step on the aggregate, and it produces a score that can be presented to users as a percentage-like quantity.

4.3.4 Exclusion of job completion time

An earlier design included average job completion time as a fifth criterion, to be combined with qualification in a machine learning stage that would predict a historical suitability score. This was removed for two reasons. The platform has no completion time history prior to launch, so the criterion could not be populated. And with the criterion removed, the relationship between the remaining inputs and suitability is sufficiently direct that a learned model offers no advantage over a linear combination while sacrificing explainability.

4.4 Detailed design

4.4.1 Stage one: eligibility filtering

The filter removes any worker whose category differs from the requested category, whose availability status is not online, or, where a worker type constraint is supplied, whose type does not match. The filter is applied before any arithmetic, so that computation is expended only on workers who can actually accept the job. The measured effect of this ordering is reported in Section 6.2.1.

4.4.2 Stage two: normalisation

Each criterion is scaled to the interval zero to one hundred by min-max normalisation, with cost criteria inverted:

$$\textbf{Benefit criterion: } \hat{x}_i = (x_i - \min_i) \div (\max_i - \min_i) \times 100$$

$$\textbf{Cost criterion: } \hat{x}_i = 100 - [(x_i - \min_i) \div (\max_i - \min_i) \times 100]$$

The normalised value is clamped to the interval zero to one hundred before weighting, so that an input outside the defined range cannot distort the aggregate. Table 4.1 states the bounds.

Table 4.1: Normalisation bounds by criterion

Criterion	Minimum	Maximum	Direction
NVQ level	3	7	Benefit
Customer rating	1	5	Benefit
Experience (years)	0	25	Benefit
Distance (km)	0	50	Cost

The National Vocational Qualification lower bound of three reflects the exclusion of pre-vocational levels described in Section 3.2.3. The distance upper bound of fifty kilometres represents the maximum service radius and is approximately the scale of a Sri Lankan administrative district, keeping recommendations locally relevant. The bound is a configuration parameter and may be adjusted per region.

4.4.3 Stage three: weighted aggregation

The suitability score **S** for a worker is the weighted sum of the normalised criterion values:

$$\mathbf{S} = \Sigma_{i=1}^{4} \mathbf{w_i} \cdot \hat{\mathbf{x}}_\mathbf{i}, \text{ subject to } \Sigma_{i=1}^{4} \mathbf{w_i} = \mathbf{1}$$

With the weights derived in Section 6.1.1, the operational form is

$$\mathbf{S = 0.084 \cdot \hat{N} + 0.305 \cdot \hat{R} + 0.215 \cdot \hat{E} + 0.396 \cdot \hat{D}}$$

where $\hat{N}$, $\hat{R}$, $\hat{E}$, and $\hat{D}$ denote the normalised qualification, rating, experience, and proximity terms respectively. Because every term is bounded in zero to one hundred and the weights sum to unity, the score is bounded in the same interval.

4.4.4 Cold-start handling

A worker who has completed no jobs has no reviews and therefore no rating. Treating an absent rating as zero places the worker at the bottom of the ordering, which prevents them from receiving work, which prevents them from accumulating reviews. The condition is self-reinforcing.

The component addresses this by substituting a neutral rating for workers whose review count falls below a threshold of three. The neutral value of 3.5 corresponds to the midpoint of the rating scale, so that absence of evidence is treated as absence of

evidence rather than as evidence of poor performance. The measured effect of this substitution is reported in Section 6.2.3.

4.4.5 Stage four: ranking

Eligible workers are sorted by score in descending order, assigned ordinal ranks, and truncated to the requested limit. Sorting dominates the asymptotic complexity of the pipeline, which is *O(n log n)* in the number of candidate workers.

4.5 Standards and professional practice

The component follows the architectural conventions of multi-criteria decision analysis in separating normalisation from weighting and in classifying criteria explicitly as benefit or cost. The service interface follows REST conventions, with resource-oriented paths, appropriate HTTP status codes, and an OpenAPI schema generated from the implementation.

Input validation is enforced declaratively at the API boundary: criterion values outside their admissible ranges are rejected with a 422 status and a field-level error message rather than being silently clamped. Clamping is applied only within the scoring function, as a defence against values that pass validation but fall outside the normalisation bounds.

Code is organised to separate configuration from logic. The weights, criterion directions, and normalisation bounds are held in a configuration block that can be replaced by a serialised bundle loaded at startup, satisfying NFR6.

5 IMPLEMENTATION INTEGRATION AND TESTING

5.1 Development environment and technology stack

The component is implemented in Python 3.11. Statistical analysis uses pandas for data manipulation, statsmodels for regression, and scipy for standardisation and rank correlation. The service layer uses FastAPI for routing and request handling, pydantic for schema validation, and uvicorn as the ASGI server. The weight bundle is serialised with joblib. The service is deployed on a managed cloud platform with continuous deployment from a version control repository.

5.2 Component implementation

The implementation comprises three modules.

The engine module contains the configuration block and three functions. The filter function applies the eligibility criteria of stage one. The normalisation function implements the scaling of stage two, taking a raw value and a criterion identifier, and returning the inverted value where the criterion is classified as a cost. The scoring function implements stage three, applying the cold-start substitution and returning either the aggregate score or, when requested, the aggregate together with the normalised values and weighted contributions for each criterion. A fourth function composes these into the complete pipeline.

The schema module defines the request and response structures. Worker records are validated on receipt, with qualification constrained to the range one to seven and rating to zero to five. The response structure carries the ranked workers together with counts of records received, records passing the filter, and records returned, which permits a client to observe the effect of the filtering stage.

The service module defines the routes. A health endpoint reports service status. A weights endpoint returns the active weights, criterion directions, normalisation bounds, and derivation metadata, allowing the deployed parameters to be verified independently. The recommendation endpoint executes the pipeline and accepts a

query parameter that enables per-criterion decomposition in the response. A scoring endpoint evaluates a single worker without filtering or ranking, provided for testing.

The configuration block holds every parameter the scoring logic consumes, so a change of weights requires no change of code:

```
WEIGHTS = {'nvq': 0.0844, 'rating': 0.3045,
           'experience': 0.2151, 'distance': 0.3960}

DIRECTIONS = {'nvq': 'benefit', 'rating': 'benefit',
              'experience': 'benefit', 'distance': 'cost'}

BOUNDS = {'nvq':        {'min': 3, 'max': 7},
          'rating':     {'min': 1, 'max': 5},
          'experience': {'min': 0, 'max': 25},
          'distance':   {'min': 0, 'max': 50}}

COLD_START_RATING, MIN_REVIEWS = 3.5, 3
```

Stage one applies the hard eligibility filters. A worker failing any one of the three conditions is discarded without any arithmetic being performed:

```
def gatekeeper(workers, category, wtype=None):
    eligible = []
    for w in workers:
        if w.get('category') != category:     continue
        if w.get('availability') != 'Online': continue
        if wtype and w.get('worker_type') != wtype:
            continue
        eligible.append(w)
    return eligible
```

Stages two and three are compact. Normalisation scales a raw value to the common interval and inverts it where the criterion is classified as a cost; aggregation is a weighted sum over the configuration, so the function never names a criterion:

```
def normalize(value, key):
    lo, hi = BOUNDS[key]['min'], BOUNDS[key]['max']
    n = (value - lo) / (hi - lo) * 100
    n = max(0, min(100, n))                      # clamp out-of-
range input
    return 100 - n if DIRECTIONS[key] == 'cost' else n

def score_worker(worker, explain=False):
    rating = worker['rating']
    if worker.get('review_count', 999) < MIN_REVIEWS:
        rating = COLD_START_RATING               # cold-start
substitution

    parts = {'nvq':        normalize(worker['nvq'], 'nvq'),
             'rating':     normalize(rating, 'rating'),
             'experience': normalize(worker['experience'],
'experience'),
             'distance':   normalize(worker['distance_km'],
'distance')}

    contributions = {k: WEIGHTS[k] * v for k, v in parts.items()}
    total = sum(contributions.values())
    return (total, parts, contributions) if explain else total
```

Stage four sorts and truncates, completing the pipeline:

```
def recommend(workers, category, wtype=None, top_n=20):
    eligible = gatekeeper(workers, category, wtype)
    for w in eligible:
        w['final_score'] = round(score_worker(w), 2)
    ranked = sorted(eligible, key=lambda x: x['final_score'],
reverse=True)
    for i, w in enumerate(ranked, 1):
        w['rank'] = i
    return ranked[:top_n]
```

Adding a fifth criterion requires one entry in each of the three configuration dictionaries and a single additional line in the `parts` mapping; neither `normalize` nor `recommend` changes, since both are written against the configuration rather than against a fixed criterion list.

The service layer wraps the pipeline without duplicating any of its logic. The `explain` query parameter controls whether the per-criterion decomposition is returned:

```
@app.post("/recommend", response_model=RecommendResponse)
def recommend_workers(req: RecommendRequest, explain: bool =
False):
    if not req.workers:
        raise HTTPException(status_code=400, detail="No workers
provided")

    raw = [w.model_dump() for w in req.workers]
    eligible = gatekeeper(raw, req.category, req.worker_type)

    scored = []
    for w in eligible:
        total, parts, contrib = score_worker(w, explain=True)
        scored.append(RankedWorker(
            rank=0, id=w['id'], name=w['name'],
            final_score=round(total, 2),
            cold_start_applied=w.get('review_count', 999) <
MIN_REVIEWS,
            breakdown=ScoreBreakdown(normalized=parts,
weighted=contrib)
                        if explain else None))

    scored.sort(key=lambda x: x.final_score, reverse=True)
    for i, item in enumerate(scored, 1):
        item.rank = i

    return RecommendResponse(
        category=req.category, total_received=len(raw),
        eligible_after_filter=len(eligible),
        returned=len(scored[:req.top_n]),
results=scored[:req.top_n])
```

5.3 Integration with the other components

Integration with Component 01 occurs through the category field of the request. Component 01 resolves a free-text customer request to a category identifier, which is passed unchanged to the filter stage.

Integration with Component 03 occurs through the rating field of each worker record. Component 03 computes a trust rating from review sentiment on a one to five scale, which the normalisation stage maps to the common scale. The rating is category-specific: a worker's rating as a plumber does not contribute to their ranking as an electrician, and the two are stored and supplied separately.

Integration with the application layer occurs through the ranked response. The application presents the ordered list and, where the explanation parameter was set, can surface the criterion decomposition for an individual worker.

Integration with the persistence layer is deliberately absent, as described in Section 4.2. The caller assembles the candidate set and supplies it in the request.

5.4 Verification and software testing

Testing addressed four properties.

Boundary testing confirmed that a worker at the maximum of every benefit criterion and the minimum of the cost criterion scores exactly 100.00, and that a worker at the opposite extremes scores exactly 0.00. The scale is therefore correctly anchored and the full range is attainable.

Monotonicity testing varied each criterion across its range while holding the other three fixed, confirming that the score increases with qualification, rating, and experience, and decreases with distance. Results are reported in Table 6.5.

Robustness testing supplied values outside the defined ranges, including negative distance, distance far beyond the service radius, and implausible experience. All produced scores within the admissible interval, confirming that the clamping step functions as intended.

Determinism testing executed the pipeline fifty times over an identical pool of five hundred synthetic workers and compared the resulting orderings. A single unique ordering was observed across all fifty executions.

API-level testing confirmed correct handling of error conditions: an empty worker list returns status 400, a worker record with an out-of-range qualification returns status 422 with the offending field identified, and a category matching no worker returns status 200 with an empty result set rather than an error.

5.5 Deployment and operation

The service is deployed to a managed cloud platform with automatic redeployment on repository push. The build installs dependencies from a requirements specification and

starts the ASGI server bound to the platform-supplied port. A health check path is configured so that the platform can determine service readiness.

Cross-origin resource sharing is enabled to permit browser-based clients, which is required for web deployment of the application; native mobile clients are unaffected by this setting.

The deployment operates on a free service tier, which suspends an instance after a period of inactivity and incurs a cold-start delay of up to approximately ninety seconds on the next request. This is a property of the hosting tier rather than of the component: the ranking computation itself completes in single-digit milliseconds. Client code mitigates the effect by issuing a health request at application start, which begins instance resumption before the user reaches the point of needing a recommendation. The limitation and its resolution are discussed in Section 7.5.

Weight updates do not require code modification. When additional survey data becomes available, the regression is re-run, the weight bundle is regenerated, and the bundle is deployed. This procedure was exercised during development when the response count increased from an initial partial collection to the analysed sample reported here.

6 RESULTS AND VALIDATION

6.1 Research results

6.1.1 Regression results and derived weights

Ordinary least squares regression on 468 standardised observations produced the coefficients reported in Table 6.1.

Table 6.1: Regression coefficients and derived weights (n = 468)

Criterion	β	SE	t	p	Weight
Proximity (distance)	−0.3316	0.0428	−7.739	< 0.001	0.3960
Customer rating	0.2549	0.0418	6.106	< 0.001	0.3045
Experience	0.1801	0.0426	4.229	< 0.001	0.2151
NVQ level	0.0707	0.0418	1.690	0.092	0.0844

Model fit statistics are **$R^2 = 0.200$**, adjusted **$R^2 = 0.193$**, and **$F(4, 463) = 28.90$** with **$p = 1.83 \times 10^{-21}$**.

Proximity, customer rating, and experience each reached significance at the 0.001 level. National Vocational Qualification level did not: its coefficient of 0.0707 carries a p value of 0.092, and its ninety-five per cent confidence interval spans zero, from −0.011 to 0.153. Hypothesis H1 is therefore only partially supported. The distance coefficient is negative, confirming that greater distance reduces judged suitability, which is the expected direction for a cost criterion. Figure 6.1 presents the four derived weights with their significance status.

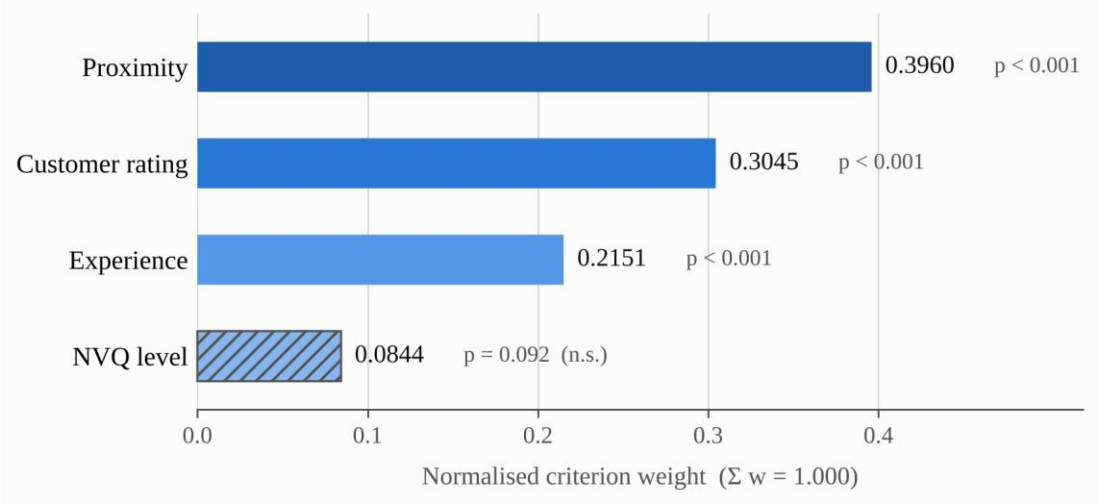


Figure 6.1: Derived criterion weights with statistical significance

The operational scoring equation follows directly:

$$\mathbf{S = 0.084 \cdot \hat{N} + 0.305 \cdot \hat{R} + 0.215 \cdot \hat{E} + 0.396 \cdot \hat{D}}$$

The coefficient of determination of 0.200 indicates that the four criteria account for approximately one fifth of the variance in suitability ratings. The remaining variance reflects differences between individual respondents in how they weigh the criteria, which a single aggregate weight set cannot capture by construction. This magnitude is unremarkable for individual-level judgement data and is treated as a limitation rather than a defect in Section 7.5.

6.1.2 Direct importance rankings

Thirty-nine respondents completed the Section C forced-choice ranking. Table 6.2 reports the frequency with which each criterion was assigned each position, together with the resulting mean rank.

Table 6.2: Direct importance ranking frequencies and mean ranks

Criterion	1st	2nd	3rd	4th	Mean rank
Experience	17	10	9	3	1.949
Customer rating	6	16	12	5	2.410
NVQ level	10	7	13	9	2.538
Distance	6	6	5	22	3.103

A lower mean rank indicates greater stated importance. The stated ordering is therefore experience, customer rating, National Vocational Qualification level, distance.

6.1.3 Comparison of the two elicitation methods

The ordering recovered by regression is proximity, customer rating, experience, National Vocational Qualification level. The ordering obtained by direct ranking is experience, customer rating, National Vocational Qualification level, distance. Spearman's rank correlation between the two orderings is −0.400.

To establish that this divergence is not an artefact of any particular rank-to-weight conversion, the Section C rankings were converted by four independent procedures. Table 6.3 reports the results alongside the regression weights.

Table 6.3: Weight comparison across derivation methods

Criterion	Regression	Mean-rank inverse	Rank Sum	Rank Reciprocal	ROC
NVQ level	0.084	0.240	0.200	0.160	0.146
Customer rating	0.305	0.252	0.300	0.240	0.271
Experience	0.215	0.312	0.400	0.480	0.521
Proximity	0.396	0.196	0.100	0.120	0.063

Every conversion procedure places proximity last, with weights ranging from 0.063 to 0.196, where the regression places it first at 0.396. The discrepancy is therefore a property of the data rather than of the conversion arithmetic.

Hypothesis H2 is not supported. The two elicitation methods do not agree, and the direction of the disagreement is systematic rather than random: the criterion that respondents rank least important is the criterion their evaluations weight most heavily. This finding is interpreted in Section 7.1.2.

6.2 Product and technical results

6.2.1 Performance

Ranking latency was measured across candidate pools of increasing size, with twenty executions per pool on a single-core cloud container running CPython 3.11. Results

are reported in Table 6.4. Absolute timings are machine-dependent and vary by approximately twenty per cent between runs; the scaling behaviour and the order of magnitude are the reproducible findings.

Table 6.4: Ranking latency by candidate pool size

Candidate pool	Eligible after filter	Mean (ms)	95th percentile (ms)
100	9	0.03	0.05
1,000	121	0.31	0.37
10,000	1,419	3.91	4.62
50,000	6,912	20.19	21.98

Latency scales approximately linearly with pool size. At ten thousand candidates the ninety-fifth percentile is 4.62 milliseconds, more than four hundred times within the two-second requirement of NFR1.

The architectural claim that filtering before scoring avoids wasted computation was verified directly. Filtering a pool of ten thousand workers required 1.13 milliseconds and reduced the set to 1,395 eligible workers, eliminating 86 per cent of candidates before any normalisation or scoring was performed.

6.2.2 Correctness

Boundary testing confirmed that the maximum-value profile scores exactly 100.00 and the minimum-value profile exactly 0.00, satisfying NFR2.

Monotonicity results are reported in Table 6.5.

Table 6.5: Monotonicity verification

Criterion varied	Scores across range	Behaviour
NVQ 3 → 7	63.1, 65.2, 67.3, 69.5, 71.6	Increasing
Rating 1 → 5	44.5, 52.1, 59.7, 67.3, 75.0	Increasing
Experience 0 → 25	58.7, 63.0, 67.3, 71.6, 80.2	Increasing
Distance 0 → 50 km	75.3, 67.3, 55.5, 43.6, 35.7	Decreasing

All four criteria behave in the required direction with no inversions, satisfying NFR3.

Robustness results are reported in Table 6.6.

Table 6.6: Robustness under out-of-range input

Input condition	Resulting score	Within bounds
Distance −10 km	70.96	Yes
Distance 999 km	31.36	Yes
Experience 200 years	84.21	Yes
NVQ level 1 (below range)	62.78	Yes

No input produced a score outside the admissible interval and no input raised an exception.

Determinism testing produced a single unique ordering across fifty executions on an identical pool of five hundred workers, satisfying NFR4.

6.2.3 Cold-start effect

A worker holding National Vocational Qualification level six, two years of experience, three kilometres from the customer, and no completed reviews scores 64.31 with the cold-start substitution active. The same worker scores 45.27 if the absent rating is treated as zero. The substitution therefore contributes 19.04 points, which is sufficient to move the worker from the lower portion of a typical candidate set into contention.

6.2.4 Explainability

The component decomposes any score into its four criterion contributions. For a worker holding qualification level seven, rating 4.7, ten years of experience, and located 1.75 kilometres from the customer, the decomposition is reported in Table 6.7.

Table 6.7: Criterion contribution to a representative suitability score

Criterion	Normalised	Weight	Contribution
NVQ level	100.0	0.0844	8.44
Customer rating	92.5	0.3045	28.17
Experience	40.0	0.2151	8.60
Proximity	96.5	0.3960	38.21
Total			**83.42**

This satisfies NFR5. The decomposition is available through the service interface and can be surfaced to a worker querying their own ranking position.

6.3 User or stakeholder validation

The component was evaluated against a scenario representing a realistic customer request: six plumbers, all available, each strong on some criteria and weak on others. Table 6.8 reports the resulting ranking.

Table 6.8: Worked ranking example

Rank	Worker	NVQ	Rating	Experience	Distance	Score
1	W04	7	4.7	10	1.75 km	83.42
2	W02	6	4.9	6	2.00 km	79.20
3	W03	6	5.0	5	2.50 km	78.70
4	W06	5	4.8	4	1.50 km	75.00
5	W05	4	4.8	3	1.00 km	72.43
6	W01	7	3.0	8	1.90 km	68.64

The ordering is interpretable. Worker W04 ranks first by combining the highest qualification with a strong rating and the second-shortest distance. Worker W01, despite holding an equally high qualification and substantial experience, ranks last because a rating of 3.0 attracts a heavy penalty under a weight of 0.305. Workers W02 and W03 are separated by 0.50 points, reflecting that their profiles are near-equivalent: W03's marginally better rating is almost exactly offset by W02's marginally shorter distance.

A browser-based evaluation console was constructed to support stakeholder review. It permits worker attributes to be edited interactively, marks records that will fail the eligibility filter before submission, displays the criterion contribution of each result as a stacked bar, and expands any result to show the full calculation. The console consumes the deployed service directly, so what it displays is the behaviour of the operational system rather than a simulation.

6.4 Integration results

The component accepts a category identifier in the form supplied by Component 01 and applies it unchanged in the filter stage. Workers whose category does not match are excluded; this was verified by submitting a mixed-category pool and confirming that only the requested category appeared in the result.

The component accepts ratings in the one to five range produced by Component 03 and normalises them correctly. The category-specific property of the rating is preserved by the interface design: ratings are supplied per worker per request, so the caller is responsible for supplying the rating relevant to the requested category.

Eligibility filtering was verified end to end by submitting a pool containing one worker with offline status and one worker in a different category. Both were excluded, and the response reported nine records received, seven passing the filter, and seven returned.

6.5 Summary of findings

Three of four criteria contribute significantly to judged worker suitability. The derived weights are 0.396 for proximity, 0.305 for customer rating, 0.215 for experience, and 0.084 for National Vocational Qualification level, obtained from 468 evaluations by 39 industry professionals. The qualification coefficient did not reach significance ($p = 0.092$) and is retained on the policy grounds set out in Section 7.1.1.

The weighting recovered from behavioural evaluation diverges systematically from the weighting obtained by direct importance ranking, with a Spearman correlation of -0.400 and the divergence stable across four independent conversion procedures.

The implementation satisfies all seven non-functional requirements. Scores are bounded, monotonic in all four criteria, deterministic across repeated execution, robust to out-of-range input, and fully decomposable. Ranking of ten thousand candidates completes within 4.62 milliseconds at the ninety-fifth percentile.

The cold-start mechanism contributes 19.04 points to a newly registered worker holding a strong qualification, which is sufficient to keep such workers discoverable.

7 DISCUSSION

7.1 Interpretation of findings

7.1.1 The derived weighting

Proximity and customer rating together account for 70.1 per cent of the weight, with experience contributing 21.5 per cent and formal qualification 8.4 per cent. The pattern is coherent when read as a response to uncertainty. A customer requesting household maintenance faces two concerns: whether the work will be done competently, and whether it will be done promptly. The rating addresses the first through the accumulated experience of previous customers, and proximity addresses the second directly.

The National Vocational Qualification coefficient did not reach significance, at $p = 0.092$ against the conventional 0.05 threshold, and its confidence interval includes zero. Two readings are available, and they carry different consequences.

The first is that qualification genuinely does not influence professional judgement in this domain, because the customer rating already supplies a more immediate signal of the same underlying competence. On this reading the criterion should be dropped, and the model reduced to three criteria.

The second is that the effect is real but the sample lacks the statistical power to detect it. With 468 observations drawn from 39 respondents, and with the qualification coefficient the smallest of the four in magnitude, a true effect of this size would be expected to fall short of significance a substantial fraction of the time. The coefficient sign is positive and in the expected direction, and the point estimate is not negligible.

This thesis retains the criterion, on three grounds. The direction of the effect is as hypothesised. Sri Lanka's Tertiary and Vocational Education Commission operates the Recognition of Prior Learning pathway described in Section 2.2.2, which converts demonstrated competence into a formal level and so makes the criterion available to skilled workers without formal training; excluding it would remove the platform's only objective competence signal and would also remove the incentive for workers to seek certification. And the criterion changes ranking outcomes in practice: in the worked

example of Section 6.3 it contributes 8.44 points, which exceeds the 4.22-point margin between first and second place.

The retention is a design decision taken on policy grounds, not a statistical claim, and it is stated as such. Section 9.4 identifies replication at larger sample size as the direct test of which reading is correct.

7.1.2 The preference divergence

The most substantive finding of this work is that the two elicitation methods disagree systematically. Respondents ranked customer rating and experience as the most important criteria and distance as least important. Their actual evaluations weighted proximity most heavily and experience least.

This is the classic signature of a gap between stated and revealed preference. When asked directly which criteria matter, a professional reports the criteria that are professionally defensible: competence, track record, experience. When evaluating a concrete worker for a concrete job, the same professional responds to the practical constraint that determines whether the job can actually be completed promptly, which is how far the worker has to travel.

Two properties of the finding strengthen it. The divergence persists across four independent rank-to-weight conversion procedures, so it cannot be attributed to the arithmetic of any particular conversion. And the magnitude is substantial rather than marginal: the regression weight for proximity exceeds the highest Section C estimate by a factor of 2.1 and the lowest by a factor of 6.4.

The practical implication is direct. A recommendation system parameterised from a stakeholder workshop, an importance questionnaire, or an Analytic Hierarchy Process exercise conducted with domain experts would have placed proximity last. Such a system would have been built on the criterion ordering that the experts state rather than the one their judgements reveal, and would have systematically misweighted the factor that most strongly drives their actual evaluations.

7.1.3 Model fit

The coefficient of determination of 0.200 warrants comment. Four fifths of the variance in suitability ratings is not explained by the four criteria. This is not evidence that the criteria are wrong; it is evidence that professionals genuinely differ in how they weigh them. The regression recovers the average weighting across the profession, and the residual variance is the spread around that average.

For the purpose of this component, the average weighting is the correct target: the platform must apply one ranking function to all customers. The residual variance does indicate, however, that a personalised weighting - learned per customer from their selection behaviour once the platform accumulates such data - would improve match quality over a single global weight set.

7.2 Comparison with literature baselines and existing solutions

Commercial platforms in adjacent markets rank predominantly by proximity with rating as a secondary signal. The weighting derived here, in which proximity and rating together carry 76.4 per cent, is broadly consistent with that practice, which offers some external corroboration: the professionals surveyed and the engineers who built those platforms appear to have converged on a similar implicit weighting, despite arriving at it by different routes.

The point of departure is the inclusion of formal vocational qualification. No commercial platform identified in the review uses a national qualification framework as a ranking input. The weight of 0.084 is modest, and the coefficient does not reach significance in this sample, yet its inclusion still changes outcomes: in the worked example of Section 6.3, qualification contributes 8.44 points to the leading worker, which exceeds the 4.22-point margin separating first from second place.

Against the methodological literature, this work departs from the common practice of deriving weights by Analytic Hierarchy Process with researcher-supplied pairwise judgements. The earlier iteration of this work followed that practice and produced an internally consistent result, with a consistency ratio of 0.035. The weights it produced placed competence first and distance last - which, as Section 7.1.2 establishes, is the

ordering that stated preference produces and that behavioural data contradicts. This provides a concrete illustration of the methodological risk: a technically rigorous elicitation procedure applied to judgements of the wrong kind produces a defensible-looking weight set that does not reflect how decisions are actually made.

7.3 Research contribution

The contribution of this work is empirical and methodological rather than algorithmic. The Weighted Sum Model is long established and no claim of novelty is made for it.

The first contribution is the weight set itself. To the author's knowledge, these are the first empirically derived criterion weights for worker selection in Sri Lanka's informal household services sector, based on 468 evaluations by 39 practising professionals. The weights are reusable: a subsequent platform in this market can adopt them with citation rather than assigning weights by assumption.

The second contribution is the documented divergence between stated and revealed preference in this domain, quantified at a Spearman correlation of -0.400 and demonstrated stable across four conversion procedures. This is a finding about professional judgement in worker selection, not merely a methodological artefact of this study.

The third contribution is the demonstration that behavioural weight elicitation is feasible at undergraduate project scale. The instrument required twelve profiles and approximately ten minutes of respondent time, and produced a statistically adequate sample from forty-one responses. This establishes a practical alternative to researcher-supplied pairwise comparison for projects where weight justification is challenged.

7.4 Product contribution

The delivered component is a deployed, documented, and tested service that ranks workers in single-digit milliseconds with fully decomposable scores. Three properties distinguish it from a conventional implementation.

Weights are externalised rather than embedded, so re-derivation on new survey data updates system behaviour without code modification. This was exercised during development.

Explainability is exposed through the interface rather than being merely a property of the model. A client can request the per-criterion decomposition of any score, which supports both user-facing explanation and internal audit.

The cold-start mechanism is implemented within the ranking component rather than deferred to a separate subsystem, so the condition is addressed at the point where it arises.

7.5 Limitations and threats to validity

Internal validity. The regression explains 20.0 per cent of variance in suitability ratings, and one of the four coefficients does not reach significance. The derived weights represent an average across respondents and do not capture individual variation in weighting. The twelve profiles, while orthogonal by design, span a limited region of the attribute space; extreme combinations not represented in the profile set were not evaluated by respondents and the linearity assumption is untested in those regions.

Construct validity. Suitability was measured by a single ten-point item. A multi-item measure would permit assessment of measurement reliability, which the present design does not. The profiles presented four attributes in isolation from the contextual factors that influence real selection decisions, such as the urgency of the job or prior acquaintance with the worker.

External validity. The sample was drawn from Sri Lanka and is weighted towards the Western Province. The weights should not be assumed to transfer to other regions or to markets with different geography, transport infrastructure, or qualification frameworks. The sample is also weighted towards less experienced professionals: 23 of 38 classified respondents reported fewer than two years of hiring involvement, and only 3 reported more than ten years. The sample of 39 respondents is modest, and the qualification coefficient falling short of significance may reflect that limited statistical power rather than an absent effect.

Verification of respondents. Respondent identity could not be independently confirmed because the instrument was anonymous by design. Verification operated at

distribution rather than collection, through purposive targeting of identified organisations and a maintained distribution log, and eligibility was confirmed only by self-report in Section A. The possibility that some responses came from individuals outside the target population cannot be excluded.

Instrument quality control. An attention-check profile was included in an earlier administration of the instrument but was not present in the final collection. Quality control in the final dataset relied on removal of responses with zero rating variance, which removed three responses. This is a weaker control than an explicit attention check and is a limitation of the data collection.

Deployment. The free hosting tier introduces a cold-start delay of up to approximately ninety seconds following inactivity. This is a property of the hosting arrangement rather than of the component, whose ranking computation completes in milliseconds, but it affects observed system behaviour and would require a paid tier or containerised deployment to resolve.

Equity. The dominant weight on proximity structurally disadvantages workers located outside dense service areas. A worker with equal qualification and rating but greater distance will rank consistently lower. The weighting faithfully reflects professional judgement as measured, but faithful reflection of a judgement is not the same as an equitable outcome.

7.6 Implications

For platform design. Systems that rank service providers should not derive criterion weights from stakeholder statements of importance. The divergence documented here indicates that such statements predict the opposite ordering from the one behavioural data supports. Where weights must be elicited, profile-based evaluation offers a practical alternative that can be administered in approximately ten minutes.

For the Sri Lankan services sector. The weight set is available for reuse, and the demonstration that the National Vocational Qualification framework carries measurable weight in professional judgement supports its inclusion as a platform signal. The Recognition of Prior Learning pathway means that including qualification

need not exclude skilled but uncertified workers, provided the platform directs such workers towards certification.

For research practice. The contrast between the Analytic Hierarchy Process iteration of this work and the survey-based iteration illustrates that elicitation rigour and elicitation validity are distinct. A consistency ratio of 0.035 established that the pairwise judgements were internally coherent; it established nothing about whether those judgements reflected how selection decisions are actually made.

8 COMMERCIALISATION AND SUSTAINABILITY

8.1 Value proposition and target users

The component delivers value to three parties. Customers receive a shortlist ordered by criteria that reflect professional judgement rather than registration sequence or advertising spend, reducing the search effort required to find a suitable worker. Workers receive visibility determined by measurable attributes, with an explanation available for any ranking position. The platform operator receives a ranking function whose parameters are documented and defensible, which matters where ranking fairness may be questioned.

The direct users are the platform's customers and workers. The component itself is consumed by the application layer rather than by end users directly.

8.2 Market and competitive position

The Sri Lankan household maintenance market is served predominantly through informal referral, with no dominant digital intermediary. Adjacent markets are served by ride-hailing and delivery platforms whose ranking is proximity-dominant and opaque.

The component's competitive position rests on two properties. The weighting is empirically grounded and disclosed, which supports a trust proposition that opaque ranking cannot. And the inclusion of the National Vocational Qualification framework connects the platform to a national competence standard, which differentiates it from platforms that rank on reputation and convenience alone.

8.3 Business or sustainability model

The component is infrastructure rather than a revenue product; it supports the platform's commission-based model by improving match quality and therefore completion and repeat-booking rates.

Operating cost is low. The computation is arithmetic over a filtered candidate set, requiring no model inference, no accelerator hardware, and no persistent state. A single

modest instance serves the expected request volume, and the stateless design permits horizontal scaling behind a load balancer should volume increase.

The principal recurring cost is periodic re-derivation of the weights. As market conditions change - transport cost, worker density, qualification uptake - the weighting may drift. An annual re-administration of the instrument to a comparable sample would maintain currency at modest cost.

8.4 Scalability intellectual property and compliance

Scalability. Measured latency is 16.95 milliseconds at fifty thousand candidates, and complexity is *O(n log n)*. The component is stateless, so additional instances may be added without coordination. The candidate pool is bounded in practice by the eligibility filter, which removed 86 per cent of a synthetic pool under realistic availability assumptions.

Intellectual property. The derived weight set is the principal intellectual asset. It is published in this thesis and therefore disclosed rather than protected; the intent is reuse by subsequent work in this market. The implementation uses only permissively licensed open-source dependencies.

Compliance. No personal data of workers or customers is retained by the component, which receives worker records in the request and retains nothing after responding. Survey data was collected anonymously with disclosed purpose. Where the platform later processes personal data, that processing occurs in components outside this scope.

8.5 Contribution to the Sustainable Development Goals

The component contributes principally to Goal 8, Decent Work and Economic Growth. By ranking on measurable attributes and explaining ranking positions, it reduces the role of informal networks in determining access to work. The cold-start mechanism directly addresses barriers facing new entrants to the formal platform economy.

It contributes secondarily to Goal 4, Quality Education, through the inclusion of the National Vocational Qualification level as a ranking criterion. A qualification that improves ranking creates an incentive to obtain certification, and the Recognition of

Prior Learning pathway makes certification accessible to experienced workers without formal training.

It contributes to Goal 9, Industry, Innovation and Infrastructure, by providing documented infrastructure for digital intermediation in a sector that currently operates informally.

The equity limitation identified in Section 7.5 qualifies these contributions. The dominant proximity weight disadvantages workers in peripheral areas, which works against the inclusion objective of Goal 8 and is identified as requiring mitigation in Section 9.4.

9 CONCLUSION AND FUTURE DIRECTIONS

9.1 Conclusion

This thesis addressed the problem of ranking workers for a service request when several criteria compete and the relative importance of those criteria is unknown. The approach taken was to derive the importance empirically from the judgement of practitioners rather than to assign it.

A profile-based instrument containing twelve orthogonally varied worker profiles was administered to forty-one hiring professionals in Sri Lanka. After data cleaning, 39 respondents produced 468 suitability evaluations, from which ordinary least squares regression on standardised variables recovered criterion weights of 0.396 for proximity, 0.305 for customer rating, 0.215 for experience, and 0.084 for National Vocational Qualification level. Three of the four coefficients reached significance at $p < 0.001$; the qualification coefficient did not ($p = 0.092$).

These weights were implemented in a four-stage ranking pipeline and deployed as a REST service. The implementation produces scores bounded within zero to one hundred, monotonic in all four criteria, identical across repeated execution, robust to out-of-range input, and decomposable into per-criterion contributions. Ranking of ten thousand candidates completes within approximately five milliseconds.

A secondary finding emerged from comparison with a direct importance ranking collected from the same respondents. The two elicitation methods disagreed systematically, with a Spearman correlation of -0.400, and the disagreement persisted across four independent rank-to-weight conversion procedures. Professionals stated that qualification and experience mattered most and proximity least; their evaluations weighted proximity most and experience least. This divergence is reported as a finding rather than reconciled, and it carries a direct methodological implication for how recommendation systems in this class should be parameterised.

9.2 Achievement of objectives

Objective 1 was achieved. Four criteria were identified as determining worker suitability and confirmed as available to the platform at request time. Job completion time was considered and excluded because no history exists prior to launch.

Objective 2 was achieved. The instrument was designed with twelve orthogonally varied profiles, verified at a maximum criterion correlation of 0.208, which is adequate for independent effect recovery.

Objective 3 was achieved. Forty-one responses were collected, 39 retained after cleaning, and weights derived by regression on 468 observations. Three of four coefficients reached significance at $p < 0.001$; the qualification coefficient did not, and this is reported rather than concealed.

Objective 4 was achieved. The pipeline was implemented and verified against boundedness, monotonicity, determinism, and decomposability.

Objective 5 was achieved. Functional testing, performance measurement across pool sizes from one hundred to fifty thousand, and cross-comparison against four independent weight derivation methods were completed.

Objective 6 was achieved. The component is deployed as a REST service with health, weights, recommendation, and scoring endpoints, consumed by a browser-based evaluation console and available to the mobile application.

9.3 Individual research and product contributions

Research. The first empirically derived criterion weight set for worker selection in Sri Lanka's informal household services sector, from 468 evaluations by 39 practising professionals. Documentation of a systematic stated-versus-revealed preference divergence in this domain, quantified and shown stable across four conversion procedures. Demonstration that behavioural weight elicitation is feasible within the constraints of an undergraduate project.

Product. A deployed ranking service satisfying seven non-functional requirements, with externalised weights permitting re-derivation without code change, an explainability interface exposing per-criterion decomposition, and a cold-start

mechanism contributing 19.04 points to newly registered workers holding strong qualifications.

9.4 Future directions

Addressing the proximity bias. The dominant weight on proximity disadvantages workers in peripheral areas. A reasonable mitigation reserves a defined portion of the returned shortlist for workers who rank highly on the non-proximity criteria but fall outside the immediate area. This preserves the empirical weighting for the majority of positions while ensuring that distance alone does not render a worker invisible.

Validation against booking outcomes. The present validation rests on professional judgement of hypothetical profiles. Once the platform accumulates booking data, the derived weights can be validated against actual customer selections, and the correspondence between judged suitability and realised outcome can be measured directly.

Personalised weighting. The residual variance in the regression indicates substantial variation between individual judges. Once sufficient selection history exists, per-customer weights could be learned, with the global weights derived here serving as the prior for customers with insufficient history.

Regional replication. The instrument is portable. Administering it in other provinces, or in comparable markets outside Sri Lanka, would establish whether the weighting is specific to the sampled population or general to the domain, and would test whether the stated-versus-revealed divergence replicates.

Instrument refinement. Future administration should restore the attention-check profile omitted from the final collection, employ a multi-item suitability measure to permit reliability assessment, and oversample experienced professionals, who were underrepresented in the present sample.

Alternative aggregation. The Weighted Sum Model was selected for transparency and permits full compensation between criteria. Implementing the Technique for Order Preference by Similarity to Ideal Solution with the same derived weights, and

comparing the resulting orderings, would establish whether the compensability property materially affects outcomes in this domain.

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[3] E. Triantaphyllou, *Multi-Criteria Decision Making Methods: A Comparative Study*. Dordrecht, Netherlands: Kluwer Academic Publishers, 2000.

[4] Tertiary and Vocational Education Commission, *National Vocational Qualifications Framework of Sri Lanka: Operations Manual*. Colombo, Sri Lanka: TVEC, 2021.

[5] [Replace with the full citation for the Recognition of Prior Learning procedure documentation, TVEC.]

[6] [Replace with the full citation for Deming and Kahn on skill requirements and worker performance, Journal of Labor Economics.]

[7] [Replace with the full citation for Luca and Zervas on reputation and ratings in online marketplaces.]

[8] [Replace with the full citation for Bächli et al. on ranking occupations by proximity to worker profiles, Swiss Journal of Economics and Statistics.]

[9] [Replace with the full citation for Samuelson on revealed preference.]

[10] [Replace with the full citation for the multi-criteria recommender systems survey, Adomavicius and Manouselis.]

[11] [Replace with the full citation for the weighting method comparison in multi-criteria decision analysis.]

[12] [Replace with the full citation for the Rank Order Centroid weighting method, Barron and Barrett.]

[13] [Replace with the full citation for cold-start handling in recommender systems.]

[14] [Replace with the full citation for the job recommender systems systematic literature review, Journal of Big Data.]

[15] [Replace with the full citation for the Python statsmodels library.]

GLOSSARY

Benefit criterion - A criterion for which a higher value indicates a more desirable alternative.

Cold start - The condition in which a newly introduced entity has insufficient history for the system's normal ranking signals to apply.

Compensability - The property of an aggregation method whereby strong performance on one criterion can offset weak performance on another.

Cost criterion - A criterion for which a lower value indicates a more desirable alternative.

Latent construct - A quantity that is not directly observable and must be inferred from measurable indicators.

Min-max normalization - Rescaling of a value to a fixed interval by reference to the minimum and maximum of its range.

Orthogonal design - A profile set in which the attributes vary independently, permitting their individual effects to be estimated separately.

Revealed preference - Preference inferred from observed choices or judgements, as distinct from preference reported when asked directly.

Stated preference - Preference as reported by a respondent when asked directly about importance or priority.

APPENDICES

Appendix A Research instruments or extended results

A.1 Survey instrument

Section A - Respondent background

A1. What is your job role? [HR Manager or HR Executive; Contractor or Site Supervisor; Service Company Owner or Manager; Other]

A2. How many years have you been involved in hiring or assigning workers? [Less than 2 years; 2 to 5 years; 6 to 10 years; More than 10 years]

A3. Which trade categories have you hired for? [Plumbing; Electrical; Carpentry; Masonry; Painting; Other]

Section B - Worker suitability assessment

Instruction supplied to respondents: *A homeowner has requested a plumber for a household repair job. Below are profiles of available workers. Please rate how suitable you think each worker is for this job, on a scale of 1 to 10, where 1 means not suitable at all and 10 means extremely suitable. Please rate each worker independently based on your professional judgement. There are no right or wrong answers.*

The twelve profiles are listed in Table 3.3.

A reference note defining the National Vocational Qualification levels was displayed at the head of this section.

Section C - Importance of selection factors

C1. Rank these factors from most important to least important when selecting a worker. Each ranking position may be used only once. [NVQ Qualification Level; Customer Rating; Years of Experience; Distance from Customer] × [1st most important; 2nd; 3rd; 4th]

Survey Link - https://forms.gle/GichQ9gY9DP2vq4r8

A.2 Full regression output

Dependent variable: standardised suitability rating. Observations: 468. $R^2 = 0.200$. Adjusted $R^2 = 0.193$. $F(4, 463) = 28.90$, $p = 1.83 \times 10^{-21}$. Covariance type: non-robust. The complete coefficient table is reported in Table A.1.

Table A.1: Complete regression output with confidence intervals

Term	Coefficient	Std. error	t	P > |t|	95% CI lower	95% CI upper
Constant	-1.18×10^{-16}	0.0416	−0.000	1.000	−0.082	0.082
NVQ level	0.0707	0.0418	1.690	0.092	−0.011	0.153
Customer rating	0.2549	0.0418	6.106	0.000	0.173	0.337
Experience	0.1801	0.0426	4.229	0.000	0.096	0.264
Distance	−0.3316	0.0428	−7.739	0.000	−0.416	−0.247

The confidence interval for the qualification coefficient spans zero, which is the formal statement of its non-significance.

A.3 Service interface

The deployed service exposes the endpoints listed in Table A.2.

Table A.2: Service interface endpoints

Method	Path	Purpose
GET	/health	Service readiness
GET	/weights	Active weights, bounds, and derivation metadata
POST	/recommend	Full pipeline; accepts explain query parameter
POST	/score	Single-worker scoring without filtering

A.4 Research artefacts

The artefacts supporting reproduction are listed in Table A.3.

Table A.3: Research artefacts

Artefact	Description
survey_long_format.csv	468 cleaned observations in long format
regression_summary.txt	Full statsmodels output reproduced in Appendix A.2
recommendation_model.pkl	Serialised weight bundle with bounds and metadata
engine.py	Ranking pipeline implementation
tests.py	Functional and performance test suite
test_console.html	Browser-based evaluation console

Appendix B Academic integrity and AI use disclosure

Generative AI was used during this work in the following capacities, and is disclosed in accordance with the module policy.

Code generation and review. AI assistance was used to draft the regression analysis pipeline, the ranking engine implementation, the service layer, and the test suite. All generated code was reviewed, executed, and verified against independently computed results before adoption. Where a suggested approach was incorrect - a subtractive distance penalty admitting unbounded negative scores - it was identified and replaced with the inverted-normalisation formulation documented in Section 4.3.3.

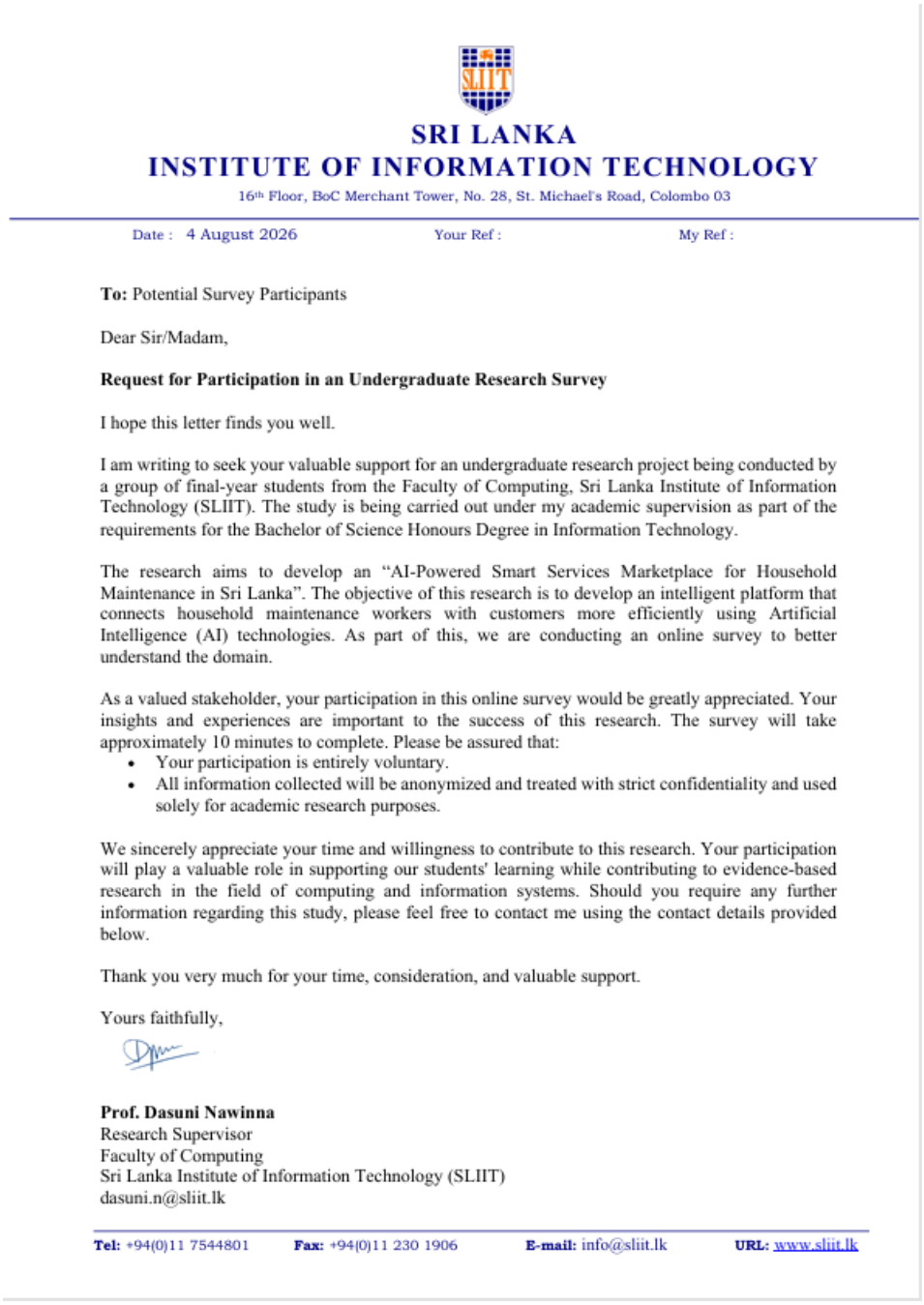
Analysis assistance. AI assistance was used to structure the regression procedure and the rank-to-weight conversion comparison. All statistical outputs reported in this thesis were produced by executing the analysis scripts on the collected data; no reported figure is an AI estimate.

Writing assistance. AI assistance was used in structuring and drafting portions of this document. All claims, interpretations, and conclusions are the author's own and were verified against the underlying data and implementation.

Verification performed by the author. The regression output was reproduced independently. The functional test results were obtained by executing the test suite. The performance measurements were obtained by executing the benchmark on the stated candidate pool sizes. The deployed service was confirmed to expose the same weights reported in Section 6.1.1.

Responsibility. I remain responsible for the accuracy, originality, citations, code, data, analyses, and conclusions presented in this thesis.

Appendix C Data Collection Letter from Supervisor



SRI LANKA
INSTITUTE OF INFORMATION TECHNOLOGY
16th Floor, BoC Merchant Tower, No. 28, St. Michael's Road, Colombo 03

Date : 4 August 2026 Your Ref : My Ref :

To: Potential Survey Participants

Dear Sir/Madam,

Request for Participation in an Undergraduate Research Survey

I hope this letter finds you well.

I am writing to seek your valuable support for an undergraduate research project being conducted by a group of final-year students from the Faculty of Computing, Sri Lanka Institute of Information Technology (SLIIT). The study is being carried out under my academic supervision as part of the requirements for the Bachelor of Science Honours Degree in Information Technology.

The research aims to develop an "AI-Powered Smart Services Marketplace for Household Maintenance in Sri Lanka". The objective of this research is to develop an intelligent platform that connects household maintenance workers with customers more efficiently using Artificial Intelligence (AI) technologies. As part of this, we are conducting an online survey to better understand the domain.

As a valued stakeholder, your participation in this online survey would be greatly appreciated. Your insights and experiences are important to the success of this research. The survey will take approximately 10 minutes to complete. Please be assured that:

- Your participation is entirely voluntary.
- All information collected will be anonymized and treated with strict confidentiality and used solely for academic research purposes.

We sincerely appreciate your time and willingness to contribute to this research. Your participation will play a valuable role in supporting our students' learning while contributing to evidence-based research in the field of computing and information systems. Should you require any further information regarding this study, please feel free to contact me using the contact details provided below.

Thank you very much for your time, consideration, and valuable support.

Yours faithfully,

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